

Key Insights

Dataset Insights

- The dataset contained **1,000 international football matches** with a balanced distribution of home wins, away wins, and draws.
- Home Wins: **424**
- Away Wins: **414**
- Draws: **162**

Team Performance Insights

- Teams with higher **Elo Ratings** generally achieved better match outcomes.
- Teams with stronger **recent form** showed a greater probability of winning matches.
- Historical **Win Percentage** proved to be a reliable indicator of long-term team strength.
- FIFA Rankings alone were not sufficient to accurately predict match outcomes.

Feature Importance Insights

The Random Forest model identified the following as the most influential factors:

1. Away Elo Rating (**11.24%**)
2. Away Form (**11.10%**)
3. Home Elo Rating (**10.71%**)
4. Home Form (**10.70%**)
5. Home Win Percentage (**9.63%**)

Model Performance Insights

- The Random Forest Classifier achieved an accuracy of **47%**.
- The model performed reasonably well when predicting home and away victories.
- Draws were the most difficult outcome to predict.
- Football remains highly unpredictable due to factors such as injuries, tactical decisions, and match-day events.

Team Strength Analysis Insights

- A Team Strength Score was developed using:
 - Elo Rating

- Recent Form
 - Win Percentage
- This score provided a comprehensive measure of overall team quality.
- Argentina consistently ranked among the strongest teams across all key performance metrics.
- Brazil, Belgium, and Austria also demonstrated strong historical performance profiles.

FIFA World Cup 2026 Prediction Insights

- Elo Rating was found to be a stronger predictor than FIFA Ranking.
- Recent Form significantly influenced match outcomes.
- Win Percentage contributed positively to team strength evaluation.
- Host Advantage and Continental Advantage showed minimal impact in the model.